

Introduction to Quantum Computing

From Classical Bits to Quantum Algorithms

2026 Edition

An Independent Modern Textbook
Research cutoff: September 2026

This textbook provides a mathematically careful yet accessible introduction to quantum computation and quantum information for readers who know basic algebra, trigonometry, and complex numbers but need not already know linear algebra, quantum mechanics, or quantum programming. It develops classical information and computation, reversible logic, single- and multi-qubit quantum theory, measurement, the Bloch sphere, linear-algebraic tools, entanglement, quantum circuits, error correction, protocols, and foundational quantum algorithms through Shor's factoring algorithm. A companion interactive website synchronizes notation, examples, and laboratory exercises with this PDF. Material labeled with the research cutoff reflects the state of the field as of September 2026 and should be re-checked as hardware and software evolve.

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How to Use This Book

Each chapter in the PDF corresponds to a section of the interactive web textbook at the project repository. When you see a **Interactive Lab** box, open the linked lab in your browser to manipulate states, run simulations, and verify calculations step by step.

Chapter 1

Classical Information and Computation

Physical systems can encode information by occupying distinguishable states. A classical bit is the simplest unit: it takes one of two values, 0 or 1. For n bits there are 2^n distinct register states.

1.1 States and Information

If each bit is independent, an n -bit register has

$$N = 2^n$$

possible configurations. For $n = 3$, the eight strings are 000, 001, ..., 111. Bits may represent numbers, text, instructions, or any digital data once an encoding convention is fixed.

Interactive Lab 1: Binary State Explorer — select n and inspect 2^n on the website.

1.2 Binary Numbers

Positional notation interprets a bit string $b_{n-1} \cdots b_0$ as

$$\sum_{j=0}^{n-1} b_j 2^j.$$

For example, $1011_2 = 1 \cdot 8 + 0 \cdot 4 + 1 \cdot 2 + 1 = 11_{10}$.

1.3 Logic Gates

Boolean variables $A, B \in \{0, 1\}$ combine through gates such as NOT ($\bar{A}$), AND ($A \wedge B$), OR ($A \vee B$), and XOR ($A \oplus B$). XOR satisfies $1 \oplus 1 = 0$ and is its own inverse.

1.4 Boolean Algebra

Identities include $A \wedge 1 = A$, $A \vee 0 = A$, and $A \wedge \bar{A} = 0$. De Morgan's laws state

$$\overline{A \wedge B} = \bar{A} \vee \bar{B}, \quad \overline{A \vee B} = \bar{A} \wedge \bar{B}.$$

1.5 Adders

A half adder outputs sum $S = A \oplus B$ and carry $C = A \wedge B$. A full adder adds an incoming carry C_{in} :

$$S = A \oplus B \oplus C_{\text{in}}, \quad C_{\text{out}} = (A \wedge B) \vee (C_{\text{in}} \wedge (A \oplus B)).$$

Ripple-carry addition propagates carries sequentially, introducing a depth cost that foreshadows circuit depth in quantum computing.

1.6 Reversible Computation

Reversible operations preserve enough information to recover inputs. NOT is reversible; AND is not. The Toffoli gate $(a, b, c) \mapsto (a, b, c \oplus ab)$ is reversible and can emulate AND when $c = 0$. Quantum evolution before measurement must be reversible (unitary).

1.7 Classical Error Correction

Three-bit repetition encoding uses $0_L = 000$ and $1_L = 111$. A single bit flip is corrected by majority vote. Parity $p = a \oplus b$ summarizes two-bit information.

1.8 Computational Complexity

Growth classes include $O(1)$, $O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$, and $O(2^n)$. The class **P** contains problems solvable in polynomial time on a deterministic classical computer. **NP** is defined decision-theoretically; quantum computers do not automatically solve NP-complete problems in polynomial time.

1.9 Turing Machines

A Turing machine reads and writes symbols on a tape, changes internal state, and moves its head according to transition rules. The Church–Turing thesis asserts that realistic classical computation is captured by this model. Quantum computation asks whether every physically realizable process is efficiently simulable classically; this motivates the class **BQP**.

Exercises

1. How many states does a 10-bit register have?
2. Simplify $\overline{A \wedge (B \vee C)}$ using De Morgan.
3. Add $1101_2 + 1011_2$ with ripple carry.

Chapter 2

The Qubit

A qubit generalizes a classical bit. Whereas a classical bit is either 0 or 1, a qubit may occupy a *superposition*—a normalized linear combination of $|0\rangle$ and $|1\rangle$.

2.1 What Is a Qubit?

A pure one-qubit state is

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1.$$

In the computational (Z) basis,

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

A qubit is *not* simply “both 0 and 1 at once” in a classical sense; amplitudes carry magnitude and phase, and interference governs observable statistics.

2.2 Complex Amplitudes

For $z = a + bi$, the conjugate is $z^* = a - bi$, magnitude $|z| = \sqrt{a^2 + b^2}$, and polar form $z = re^{i\phi}$ with Euler’s formula $e^{i\phi} = \cos \phi + i \sin \phi$.

2.3 Measurement in the Z Basis

Measuring $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$ in the Z basis yields outcome 0 with probability $|\alpha|^2$ and outcome 1 with probability $|\beta|^2$. After outcome 0, the state collapses to $|0\rangle$.

Example. For $|\psi\rangle = \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle$, we have $P(0) = 3/4$ and $P(1) = 1/4$.

2.4 Other Measurement Bases

The X basis uses $|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$ and $|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$. The Y basis uses $|i\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}$ and $|-i\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}}$. Rewriting $|\psi\rangle$ in a chosen basis and squaring coefficient magnitudes gives measurement probabilities.

2.5 Global and Relative Phase

Multiplying $|\psi\rangle$ by a global phase $e^{i\gamma}$ does not change Z-basis probabilities. Relative phase between $|0\rangle$ and $|1\rangle$ *does* matter: $|+\rangle$ and $|-\rangle$ have identical Z statistics but differ in X statistics.

2.6 The Bloch Sphere

Ignoring global phase, any pure qubit satisfies

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle,$$

with Bloch coordinates $x = \sin \theta \cos \phi$, $y = \sin \theta \sin \phi$, $z = \cos \theta$. Important states include $|0\rangle \mapsto (0, 0, 1)$, $|1\rangle \mapsto (0, 0, -1)$, $|+\rangle \mapsto (1, 0, 0)$, and $|i\rangle \mapsto (0, 1, 0)$.

2.7 One-Qubit Gates

Common gates include identity I , Pauli X, Y, Z , Hadamard H , phase S , and T . For example,

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad H|0\rangle = |+\rangle.$$

Identities include $X^2 = Y^2 = Z^2 = H^2 = I$ and $S^2 = Z$, $T^2 = S$.

Interactive Labs 4–8: Complex plane, measurement histograms, Bloch sphere, and gate explorer on the website.

Exercises

1. Normalize $(2, 1)^\top$ and compute $P(0)$.
2. Express $|-\rangle$ in the Z basis and compute $P(-)$ for $|0\rangle$.
3. Apply H then Z to $|+\rangle$ and give Bloch coordinates.

Chapter 3

Linear Algebra for Quantum Computing

Quantum states are vectors; quantum gates are matrices. This chapter introduces the Dirac notation and linear tools used throughout the book.

3.1 Kets and Bras

We write $|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ and $\langle\psi| = [\alpha^* \ \beta^*]$.

3.2 Inner Products

For $|\phi\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$ and $|\psi\rangle = \begin{bmatrix} c \\ d \end{bmatrix}$,

$$\langle\phi|\psi\rangle = a^*c + b^*d, \quad \|\psi\| = \sqrt{\langle\psi|\psi\rangle}.$$

Computational basis vectors are orthonormal: $\langle 0|0\rangle = \langle 1|1\rangle = 1$ and $\langle 0|1\rangle = 0$.

3.3 Projection and Measurement

The amplitude for outcome $|m\rangle$ is $\langle m|\psi\rangle$ and

$$P(m) = |\langle m|\psi\rangle|^2.$$

The projector is $P_m = |m\rangle\langle m|$.

3.4 Outer Products and Completeness

$|\psi\rangle\langle\phi|$ is a rank-one matrix. Completeness in the computational basis:

$$|0\rangle\langle 0| + |1\rangle\langle 1| = I.$$

3.5 Unitary Matrices

A gate U is unitary if $U^\dagger U = U U^\dagger = I$, hence $U^{-1} = U^\dagger$. Unitary evolution preserves normalization:

$$\langle \psi' | \psi' \rangle = \langle \psi | U^\dagger U | \psi \rangle = \langle \psi | \psi \rangle.$$

Interactive Lab 9: Unitary checker — enter a 2×2 matrix and verify $U^\dagger U$.

Exercises

1. Compute $\langle + | - \rangle$ and interpret orthogonality.
2. Show that H is unitary by computing $H^\dagger H$.
3. Write $|1\rangle\langle 0|$ as a matrix.

Chapter 4

Composite Quantum Systems

Multiple qubits are described by tensor products. An n -qubit computational basis has 2^n states.

4.1 Tensor Products

Two-qubit product states satisfy $|\psi\rangle \otimes |\phi\rangle$; for example $|0\rangle \otimes |1\rangle = |01\rangle$. A general two-qubit state is

$$|\psi\rangle = c_{00}|00\rangle + c_{01}|01\rangle + c_{10}|10\rangle + c_{11}|11\rangle,$$

with $\sum |c_{ij}|^2 = 1$.

4.2 Entanglement

A state is *separable* if it equals a product $|\alpha\rangle \otimes |\beta\rangle$; otherwise it is *entangled*. Bell states include

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Measuring one qubit of $|\Phi^+\rangle$ in the Z basis yields correlated outcomes; this does not enable faster-than-light signaling.

4.3 Multi-Qubit Gates

The CNOT gate acts as $|a, b\rangle \mapsto |a, a \oplus b\rangle$. Creating $|\Phi^+\rangle$ from $|00\rangle$ uses H on the first qubit followed by CNOT. SWAP exchanges qubits; Toffoli generalizes classical reversible AND.

4.4 No-Cloning Theorem

No unitary process can clone an unknown arbitrary qubit: cloning would require $\langle\psi|\phi\rangle = \langle\psi|\phi\rangle^2$ for all pairs, which fails unless states are orthogonal.

4.5 Universal Gate Sets

The set $\{H, T, \text{CNOT}\}$ is universal: any n -qubit unitary can be approximated to arbitrary precision using finitely many gates from a universal set (Solovay–Kitaev theorem, stated without proof).

Interactive Labs 10–12: Tensor products, Bell pairs, entanglement measurement.

Exercises

1. Write $|+\rangle \otimes |0\rangle$ in the four-bit computational basis.
2. Show $|\Phi^+\rangle$ is not a product state.
3. Apply CNOT to $\frac{|00\rangle+|10\rangle}{\sqrt{2}}$.

Chapter 5

Quantum Error Correction

Real devices decohere. Error correction protects *logical* qubits using redundancy in *physical* qubits.

5.1 Noise and Decoherence

Errors include bit flips (X), phase flips (Z), depolarizing noise, relaxation (T_1), and dephasing (T_2). As of the September 2026 research cutoff, large-scale fault-tolerant quantum computation remains an active engineering goal; recent experiments demonstrate improving logical error rates with surface-code scaling [1], while commercial roadmaps should be distinguished from established theorems.

5.2 Three-Qubit Bit-Flip Code

Logical encoding: $|0_L\rangle = |000\rangle$, $|1_L\rangle = |111\rangle$. An arbitrary logical qubit $\alpha|0_L\rangle + \beta|1_L\rangle$ can suffer a single X error on one wire; parity measurements reveal the syndrome without collapsing α, β .

5.3 Phase-Flip Code

Phase errors flip $|+\rangle \leftrightarrow |-\rangle$. Conjugation by H exchanges X and Z , reducing phase-flip correction to bit-flip correction in the rotated basis.

5.4 Shor Nine-Qubit Code

The Shor code combines bit- and phase-flip protection using nine physical qubits per logical qubit. Modern research emphasizes surface codes and other topological schemes with higher thresholds and more hardware-friendly layouts.

5.5 Error Mitigation vs. Correction

In the NISQ era [5], error *mitigation* techniques (zero-noise extrapolation, probabilistic error cancellation) reduce bias without full logical encoding. Mitigation is not a substitute for fault tolerance at scale.

Interactive Lab 14: Inject bit-flip errors and correct with syndromes.

Exercises

1. What syndrome indicates an X error on qubit 2 in the three-bit code?
2. Why cannot you clone $|\psi\rangle$ to protect it?
3. Distinguish physical and logical qubits in one paragraph.

Chapter 6

Quantum Circuits and Programming

Quantum algorithms are often expressed as circuits: horizontal wires are qubits, time flows left to right, boxes are gates, control dots mark controlled operations, and meters denote measurement.

6.1 The Circuit Model

An n -qubit state evolves as $|\psi_{\text{out}}\rangle = U_k \cdots U_1 |\psi_{\text{in}}\rangle$ with each U_i a unitary on some qubits. Circuit *depth* counts sequential layers; *gate count* totals operations.

6.2 Building Bell States

$$|00\rangle \xrightarrow{H_0} \xrightarrow{\text{CNOT}_{0,1}} |\Phi^+\rangle.$$

6.3 Modern Software Ecosystem (2026)

Production quantum software typically combines:

- **Circuit construction:** OpenQASM 3, Qiskit, Cirq, PennyLane, and vendor SDKs.
- **Simulation:** local state-vector simulators (exponential cost in qubits) and stabilizer simulators for Clifford circuits.
- **Execution:** cloud access to superconducting, trapped-ion, neutral-atom, and photonic backends with calibration data and error models.

The interactive website includes a drag-and-drop circuit builder with state-vector readout; no external account is required for core learning labs.

6.4 Reading Quantum Programs

A Qiskit-style Bell circuit prepares $|\Phi^+\rangle$ using `h(0)` and `cx(0,1)`. Always map API calls back to unitaries and tensor-product structure on the underlying Hilbert space.

Interactive Lab 13: Circuit Builder on the website.

Exercises

1. What is the unitary for $H \otimes I$ on two qubits?
2. Count depth of a ripple-carry adder built from Toffoli gates (qualitatively).
3. Export a Bell circuit to OpenQASM mentally: which gates appear?

Chapter 7

Entanglement and Quantum Protocols

Entanglement enables communication protocols impossible with classical bits alone, while respecting the no-signaling principle.

7.1 EPR and Hidden Variables

Entangled correlations prompted questions about local hidden variables. Quantum mechanics predicts correlations beyond any local classical model for certain measurements.

7.2 CHSH Inequality

Define correlations $E(a, b)$ for measurement settings a, b . The CHSH combination

$$S = E(a, b) + E(a, b') + E(a', b) - E(a', b') \leq 2$$

for local hidden-variable theories. Quantum mechanics permits $|S| \leq 2\sqrt{2}$ (Tsirelson bound).

7.3 Superdense Coding

Shared $|\Phi^+\rangle$ lets Alice encode two classical bits into one qubit by applying I, X, Z , or XZ before sending her qubit to Bob, who performs Bell decoding.

7.4 Quantum Teleportation

Unknown $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ transfers to Bob using shared entanglement, local gates, measurement, and two classical bits. No matter moves; Alice's original state is destroyed; signaling faster than light is impossible.

7.5 BB84 Quantum Key Distribution

Alice encodes random bits in random Z or X bases; Bob measures in random bases. Matching bases yield sifted key bits; sampling reveals eavesdropping via elevated error rates.

Interactive Labs 15–18: CHSH, superdense coding, teleportation, BB84.
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Exercises

1. After Alice measures $|\Phi^+\rangle$ and gets 1, what is Bob's qubit?
2. Why does Eve's intercept-resend attack increase the error rate?
3. Compute S for a singlet with optimal angles (sketch).

Chapter 8

Quantum Algorithms

Quantum algorithms exploit interference and phase to outperform classical strategies for specific problems.

8.1 Oracles and Phase Kickback

Standard oracle: $U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$. With $|y\rangle = |-\rangle$, phase kickback yields $(-1)^{f(x)} |x\rangle |-\rangle$, implementing a phase oracle $O_f |x\rangle = (-1)^{f(x)} |x\rangle$.

8.2 Deutsch and Deutsch–Jozsa

For $f : \{0, 1\} \rightarrow \{0, 1\}$, Deutsch’s algorithm distinguishes constant from balanced functions with one query versus two classically in the worst case. Deutsch–Jozsa generalizes to n bits with one query versus up to $2^{n-1} + 1$ classically.

8.3 Bernstein–Vazirani

Hidden string $s \in \{0, 1\}^n$ with $f(x) = s \cdot x \bmod 2$ is found with one query quantumly versus n classically.

8.4 Simon’s Algorithm

Periodic functions $f(x) = f(x \oplus s)$ yield equations $y \cdot s = 0$ over $\mathbb{F}_2$; enough samples determine s via linear algebra.

8.5 Grover Search

Unstructured search over $N = 2^n$ items uses $O(\sqrt{N})$ oracle queries via alternating oracle $O_w = I - 2|w\rangle\langle w|$ and diffusion $D = 2|s\rangle\langle s| - I$. Optimal iterations $\approx \lfloor \frac{\pi}{4} \sqrt{N} \rfloor$.

8.6 Quantum Fourier Transform

$$\text{QFT}_N |x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i x k / N} |k\rangle.$$

The QFT is unitary and efficiently implementable, but amplitudes are not all readable at once.

8.7 Quantum Phase Estimation

Given $U|u\rangle = e^{2\pi i \phi}|u\rangle$, controlled powers of U and inverse QFT estimate ϕ to m bits using m ancilla qubits.

8.8 Shor's Factoring Algorithm

Factoring reduces to finding period r of $f(x) = a^x \bmod N$. For $N = 15$, $a = 2$, we find $r = 4$, then $\gcd(2^{r/2} \pm 1, N) \in \{3, 5\}$. Quantum speedup lies in period finding; remaining steps are classical [6].

Interactive Labs 19–27: Full algorithm simulators on the website.

Exercises

1. Which of the four $n = 1$ Boolean functions is constant?
2. Run Grover on $N = 8$ with marked $|011\rangle$; how many iterations?
3. For QPE with $m = 3$, what binary fraction approximates $\phi = 1/3$?

Chapter 9

Modern Quantum Computing in 2026

This chapter summarizes topics a serious introductory student should know beyond foundational algorithms. Statements are tagged by evidence type; company roadmaps are not scientific facts.

9.1 Physical Qubit Platforms

Leading platforms include superconducting circuits, trapped ions, neutral atoms, photonics, and spin qubits. Each trades coherence time, gate fidelity, connectivity, and scalability differently. No platform has yet demonstrated fault-tolerant logical computation at scale suitable for cryptographically relevant factoring.

9.2 Logical Qubits and the Surface Code

The surface code [1] is a leading topological code with local checks on a 2D lattice. Logical error rates can fall below physical rates when sufficiently large codes are used; resource estimates for useful chemistry or factoring remain substantial.

9.3 Noise, Channels, and Mitigation

Completely positive trace-preserving maps model open systems. NISQ devices benefit from calibration, dynamical decoupling, and error mitigation; fault tolerance requires redundancy and continual syndrome extraction.

9.4 Quantum Advantage and Benchmarks

Random circuit sampling milestones [2] demonstrate quantum behavior hard to simulate classically for specific tasks; this is not universal speedup for everyday computing. Always ask: *which problem, which metric, which comparison?*

9.5 Quantum Networking and Cryptography

Quantum key distribution implements information-theoretic security assumptions under device trust models. Post-quantum *classical* cryptography addresses Shor’s threat to RSA/ECC with lattice-based and other schemes standardized separately from quantum hardware progress.

9.6 Resource Estimation and Classical Simulation

Classical simulation of n qubits requires $O(2^n)$ complex amplitudes in general; Clifford circuits simulate efficiently. Understanding this gap clarifies why variational algorithms and mitigation target near-term hardware.

9.7 Open Questions at the September 2026 Cutoff

- Fault-tolerant logical qubits at scale with practical overheads.
- Which application domains yield durable quantum advantage.
- Integration of quantum accelerators with classical HPC workflows.

Further Reading

See [4, 5, 3] and peer-reviewed surveys. Re-verify hardware claims against primary literature.

Appendix A

Gate Reference

Standard single-qubit gates:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix},$$
$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}, \quad T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}.$$

Rotation gates:

$$R_x(\theta) = e^{-i\theta X/2}, \quad R_y(\theta) = e^{-i\theta Y/2}, \quad R_z(\theta) = e^{-i\theta Z/2}.$$

Two-qubit:

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

Useful identities: $HXH = Z$, $HZH = X$, $H^2 = I$, $T^2 = S$, $S^2 = Z$.

Appendix B

Formula Sheet

Single qubit: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, $|\alpha|^2 + |\beta|^2 = 1$.

Measurement: $P(i) = |\langle i|\psi\rangle|^2$.

Bloch: $|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$.

Unitary: $U^\dagger U = I$.

Bell: $|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$.

Grover: $O_w = I - 2|w\rangle\langle w|$, $D = 2|s\rangle\langle s| - I$.

QFT: $\text{QFT}|x\rangle = \frac{1}{\sqrt{N}} \sum_k e^{2\pi i x k / N} |k\rangle$.

Shor: $a^r \equiv 1 \pmod{N}$, factors from $\gcd(a^{r/2} \pm 1, N)$.

Appendix C

Complexity Reference

Problem	Classical	Quantum
Deutsch ($n = 1$)	2 queries (worst)	1 query
Deutsch–Jozsa	$2^{n-1} + 1$	1
Bernstein–Vazirani	n	1
Grover search	$O(N)$	$O(\sqrt{N})$
Factoring (Shor)	no known poly-time	poly-time (ideal model)

BQP is the class of decision problems solvable in polynomial time with bounded error on a quantum computer. BQP is contained in PSPACE; the relationship to NP is unknown.

Caution: Different algorithms provide different advantages. Quantum computers are not universally exponentially faster.

Appendix D

Glossary

Amplitude Complex coefficient of a basis state.

Ancilla Auxiliary qubit used during a computation.

Basis Orthonormal set spanning a Hilbert space.

Bit Classical two-level information unit.

Bloch sphere Geometric picture of a pure qubit state.

BQP Bounded-error quantum polynomial time complexity class.

CNOT Controlled-NOT gate.

Decoherence Loss of quantum coherence via environment coupling.

Entanglement Non-separable quantum correlation.

Gate Reversible operation on bits or qubits.

Global phase Overall phase with no observable effect.

Hadamard Gate mapping Z and X bases.

Hilbert space Complex vector space of quantum states.

Ket Dirac vector notation $|\psi\rangle$.

Measurement Probabilistic extraction of classical outcomes.

Oracle Black-box function access in algorithms.

Phase Argument of a complex amplitude.

Qubit Two-level quantum information unit.

Superposition Linear combination of basis states.

Tensor product Composition of subsystem state spaces.

Unitary Norm-preserving reversible linear operator.

Bibliography

- [1] Rajeev Acharya et al. Suppressing quantum errors by scaling a surface code logical qubit. *Nature*, 614:676–681, 2023.
- [2] Frank Arute et al. Quantum supremacy using a programmable superconducting processor. *Nature*, 574:505–510, 2019.
- [3] IBM Quantum. Ibm quantum platform documentation, 2026. Accessed September 2026.
- [4] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information*. Cambridge University Press, 2010.
- [5] John Preskill. Quantum computing in the NISQ era and beyond. *Quantum*, 2:79, 2018.
- [6] Peter W. Shor. Algorithms for quantum computation: Discrete logarithms and factoring. *Proceedings of the 35th Annual Symposium on Foundations of Computer Science*, 1994.